
EE 591: Selected Topics of Electric Drives

Course Code	EE 591
Course Name	Selected Topics of Electric Drives
L-T-P-C	1-0-0-1
Prerequisites	EE 309 (Power Electronics), EE 201 (Electromechanics), EE508(Fundamentals of Electric Drives)
Intended for	UG/PG

Course Contents

1. Drives Characteristics (1 hour)
2. Closed Loop Design in DC Drives (2 hours)
3. Generalized Modelling of AC Machines (2 hours)
4. Field Oriented Control (FOC) of AC Machines (5 hours)
5. Modelling, Control Design and FOC of PMSM. Modelling, Control Design and FOC of SCIM. Direct Torque Control (DTC) of Induction Machine (2 hours)
6. Modulation Techniques (2 hours)
7. Other Advanced Topics in Electric Drives (if time permits)

Text Books

1. Werner Leonhard, **Control of Electrical Drives**, 3rd Edition, Springer 2001.
2. Peter Vas, **Sensorless Vector and Direct Torque Control**, Oxford, 1998.
3. R. Krishnan, **Electric Motor Drives: Modeling, Analysis, and Control**, Prentice Hall, 2001.
4. Bimal K Bose, **Modern Power Electronics and AC Drives**, Prentice Hall, 2001.